SHRI RAMDEOBABA COLLEGE OF ENGINEERING AND MANAGEMENT, NAGPUR

Department of Computer Science and Engineering (AIML)

Semester : VI

Session 2025-26

Project Seminar on

AI-DRIVEN NEUROMORPHIC DIAGNOSTICS FOR PRECISION AGRICULTURE

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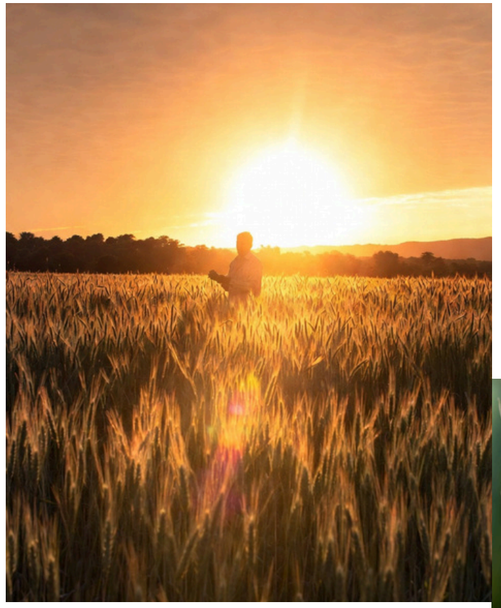
Introduction

- Wheat is one of the most important staple crops, feeding billions of people globally and playing a crucial role in India's agricultural economy.

- However, wheat production is significantly affected by diseases such as rust, leaf blight, Karnal bunt, and Fusarium head blight, which can lead to severe yield losses and economic damage.

- Traditional disease detection relies on manual visual inspection, which is slow, subjective, and often identifies infections only after visible symptoms appear.
- Moreover, existing deep learning solutions demand high computational power, making them unsuitable for low-power field devices.



Problem Definition

- Productivity is severely affected by diseases such as: Rust, Leaf blight, Karnal bunt ,Fusarium head blight
- Early detection is critical to prevent severe damage.
- Existing detection methods rely on manual visual inspection.
- Manual inspection is slow, subjective, and often unreliable.
- Diseases are usually detected only after visible symptoms appear.
- Late detection leads to delayed treatment and higher crop losses.

Literature Survey

SOURCE	YEAR	FINDINGS
YOLOv12: Attention-Centric Detectors	2025	Replaced standard CNN backbones with a pure attention design using Area Attention and a Residual Efficient Layer Aggregation Network (R-ELAN).
EBS-YOLO for Wheat FHB	2025	Developed an Edge-Optimized Bidirectional Spatial Feature Augmentation network using a Focal-Edge Selection Module (FSM) to capture indistinct lesion boundaries.
NeuEdge Framework	2025	Integrated adaptive Spiking Neural Networks (SNNs) with an Adaptive Threshold Mechanism that dynamically adjusts neuron firing based on input activity.
SNN with Bernoulli Trials	2025	Converted static leaf images into temporally encoded spike trains and integrated Bernoulli trials to regulate spike generation and suppress redundant activity.

Literature Survey

SOURCE	YEAR	FINDINGS
S2NN (Spectra-Spiking Neural Network)	2025	Built a non-destructive wheat health system using Hyperspectral Imaging and SNNs. Used membrane potential as memory to enable energy-efficient processing via sparsity.
Spiking-YOLO (Foundational)	2019	The landmark research that first successfully converted a YOLO object detection model into a Spiking Neural Network. By using channel-wise normalization, they proved that neuromorphic conversions could run on specialized chips and consume drastically less energy than standard YOLO models while maintaining object detection capabilities.
Plantix & Agrio (Existing Solutions)	2025	Commercial mobile crop doctors that allow farmers to diagnose crop damage via smartphone cameras. While successful, they rely heavily on traditional, power-hungry Artificial Neural Networks (ANNs) and often require cloud processing. This highlights the real-world need for offline, energy-efficient neuromorphic alternatives.
MDDM-WD Framework	2025	A hybrid VGG-16 and SVM approach for multi-disease wheat detection reached 97% accuracy for diseases like stripe rust and yellow dwarf.

Research Gap

Motivation

-Growing agricultural challenges:
Wheat diseases continue to cause major yield losses and economic instability for farmers, especially in developing regions where early detection infrastructure is limited.

-**Need for early and precise diagnosis:**
Traditional visual inspection detects diseases only after visible symptoms appear, leading to delayed treatment and excessive pesticide usage.

-**Demand for energy-efficient AI solutions:**
Most deep learning models are computationally intensive and unsuitable for low-power edge devices such as smartphones and drones used in field monitoring.

-**Bridging AI and neuromorphic computing for agriculture:**
Combining attention-based object detection with Spiking Neural Networks enables real-time, low-power, and scalable disease diagnostics, making advanced AI accessible to farmers.

Objectives

- To create an AI-driven neuromorphic diagnostic system called Ann Yoddha, aimed at the early detection and monitoring of wheat diseases.
- To design an attention-centric YOLOv12-based model for accurate detection of early-stage wheat disease symptoms
- To integrate Spiking Neural Networks (SNNs) to significantly reduce power consumption while preserving detection accuracy.
- To enable real-time disease diagnosis on low-power edge devices such as smartphones and UAVplatforms.
- To develop a multilingual mobile application that provides instant diagnosis and treatment recommendations
- To minimize excessive pesticide usage through precise and localized disease identification

Proposed Plan of Work

01

Stage 1

Collection and Preparation of Wheat Disease Image Data

02

Stage 2

Enhancing Dataset Robustness Through Real-World Augmentation

03

Stage 3

Development and Optimization of an Attention-Based YOLOv12-Nano Model for Early Disease Detection

04

Stage 4

Conversion of Deep Learning Model into Energy-Efficient Hybrid ANN–SNN Architecture

05

Stage 5

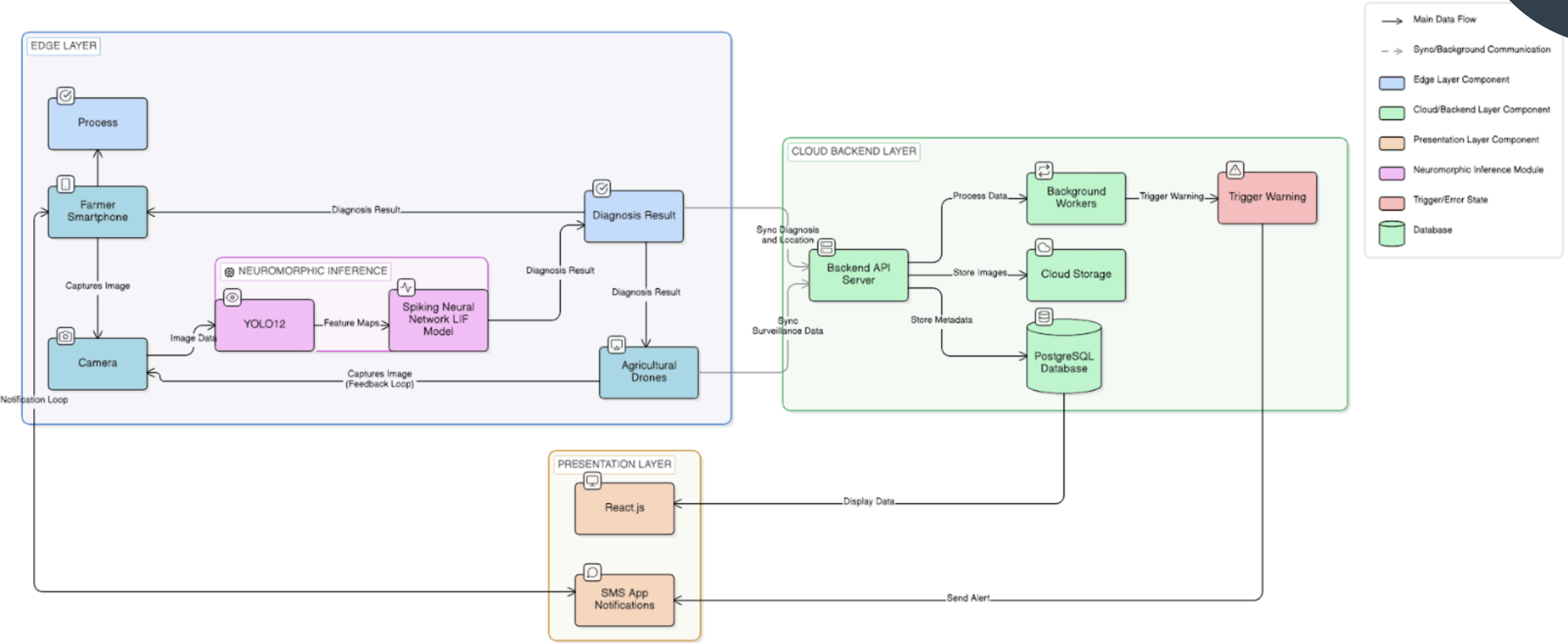
Design and Integration of Mobile Application and Web-Based Monitoring Dashboard

06

Stage 6

Real-World Field Deployment and Comprehensive Performance Evaluation

Proposed Plan of Work



Expected Results

Mobile Diagnostic App

Development of a multilingual mobile application for real-time wheat disease diagnosis.

Web Analytics Dashboard

Creation of a web-based dashboard to visualize regional disease hotspots and conduct predictive analytics.

AI Model Deployment

Provision of optimized Artificial Neural Network (ANN) and Spiking Neural Network (SNN) model files for edge deployment.

Automated Alert System

Implementation of automated diagnostic reports and outbreak alert generation.

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Thank You